

PORTFOLIO 2026

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Mechanical Engineering
& Computing Technology
at uOttawa

Co-op · Graduating April 2027 · Ottawa, ON



ABOUT ME

Building is my passion. Understanding is my method.

School design projects, household repairs, prototypes I want to test in SolidWorks - I go looking for problems worth solving. When something fails, I take it apart, find out why, and rebuild it better.

I'm competitive, and I commit fully to what I take on. Right now that means a six-axis robot arm on the bench, piano in the evenings, and training for my next fight.

Muay Thai

Scuba Diving

Volleyball

Travel

Reading



**CURTISS-
WRIGHT**

WORK

CURTISS-WRIGHT · 2025

Liquid-Cooled Chassis Redesign

I reworked the liquid cooling geometry of a ruggedized electronics part to see if cheaper machining methods could hold the same thermal performance.

MY ROLE

Redrew the cooling paths in CAD around cheaper machining processes, then ran Flotherm simulations to compare the new geometry against the original.

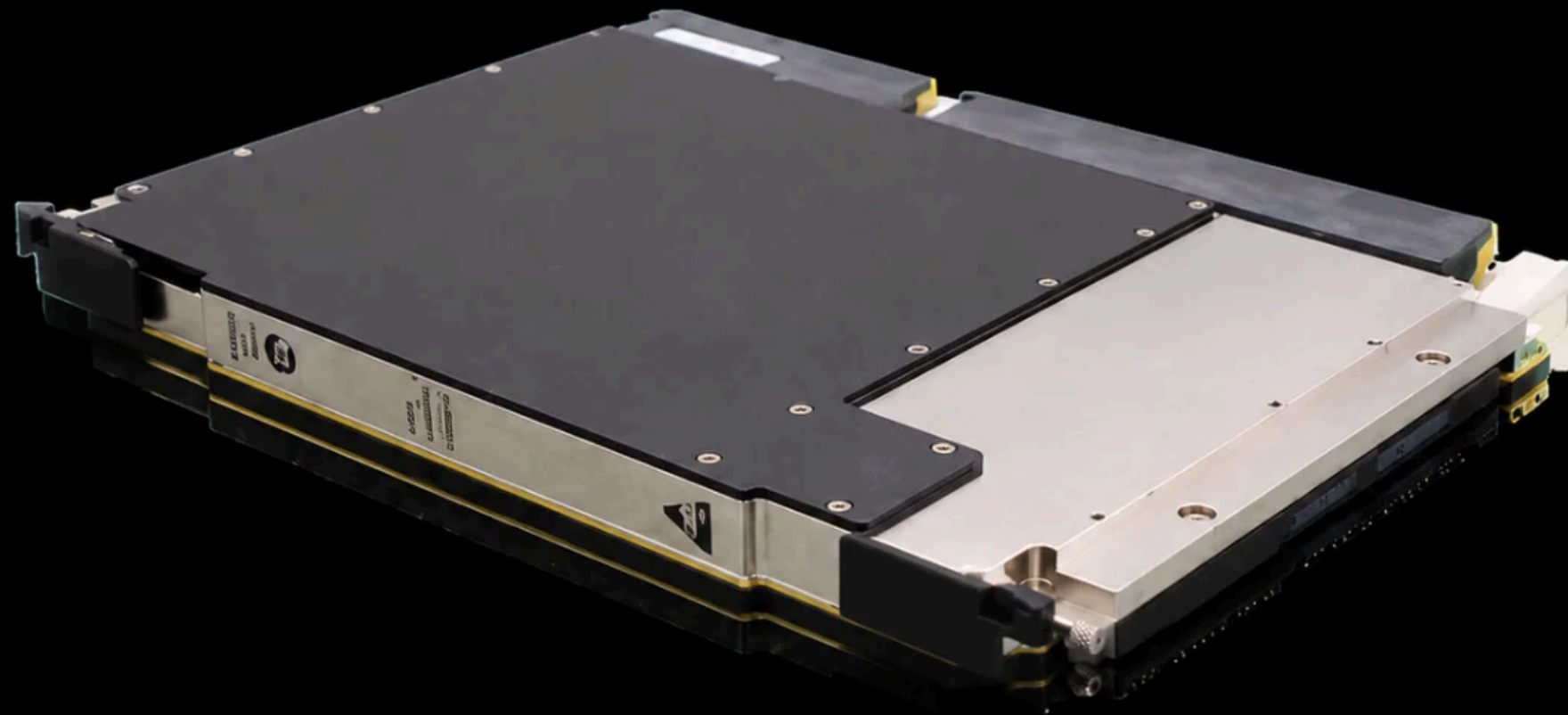
OUTCOME

A manufacturable, lower-cost layout the team could build against, alongside thermal and environmental testing on nine conduction-cooled products.

Flotherm

Thermal design

CAD



WORK CANMETENERGY · RH&P · 2025

Aspirated & Shielded RTD Sensor

Air temperature readings inside the SolarSteam enclosure were being thrown off by radiation, so I built a shielded sensor that slides up and down to sample at any height.

MY ROLE Designed the shield in SolidWorks, sized the aspiration from the literature, and ran the accuracy testing myself.

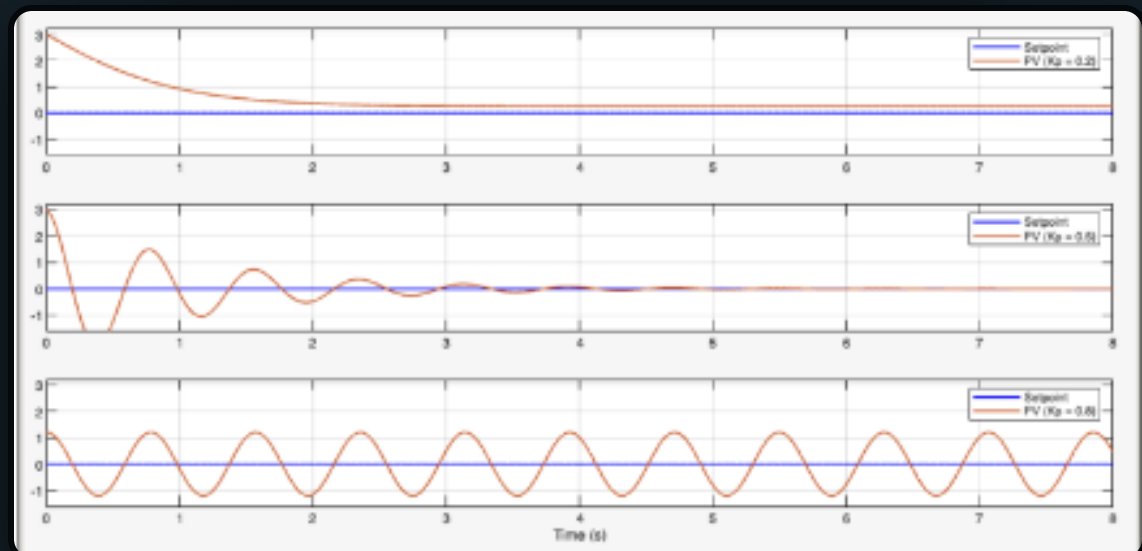
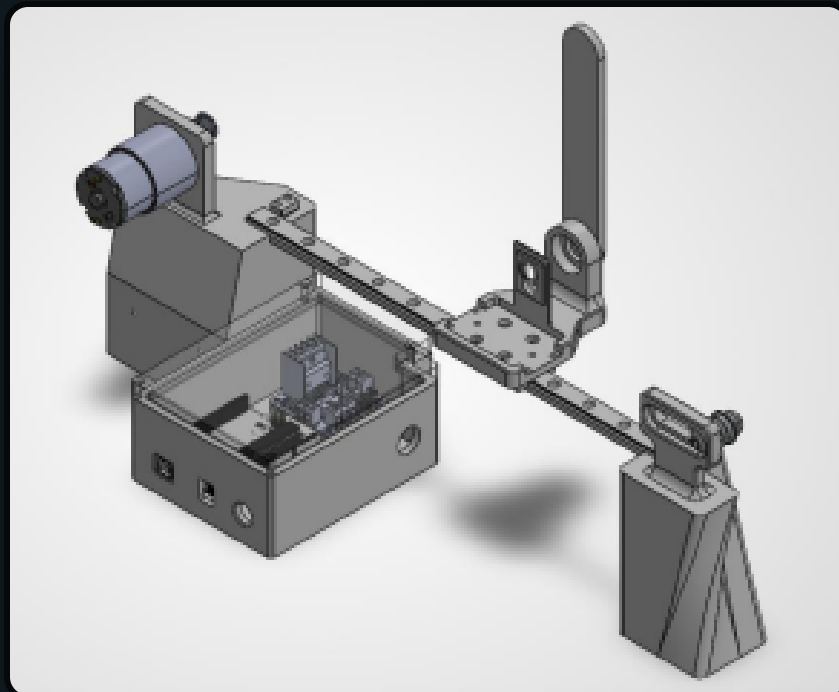
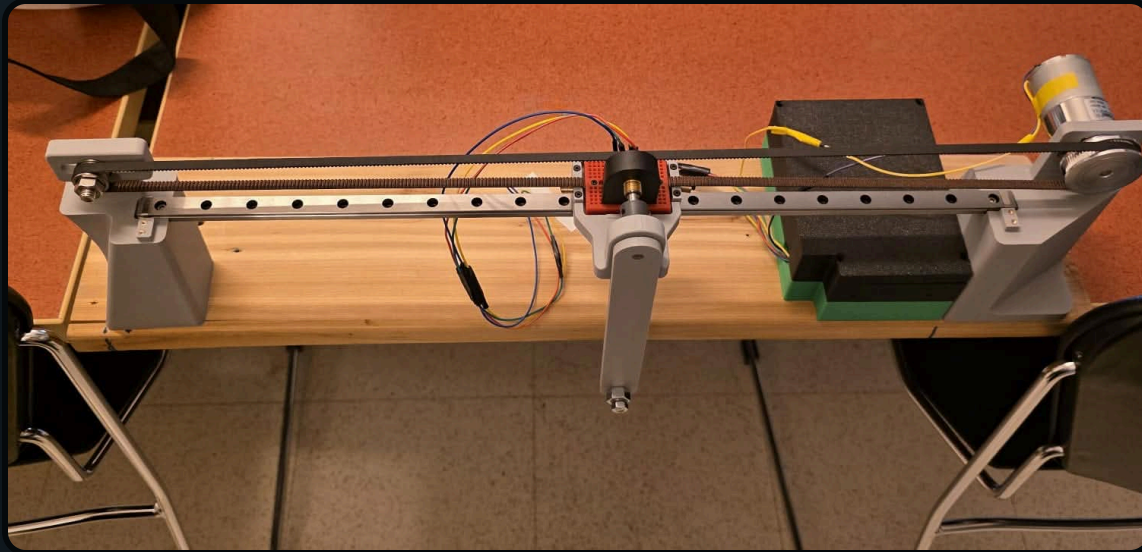
OUTCOME Live temperature accuracy improved by more than 30%, and I presented the work as a research poster.

SolidWorks

Instrumentation

Field testing





SCHOOL CONTROL SYSTEMS

Inverted Pendulum

A belt-driven cart on a 0.5 m rail that balances a rigid pendulum upright, with a rotary encoder reading the angle and nothing telling it where the cart is.

MY ROLE Designed every 3D model on the rig, Sourced all parts, optimized the electronics layout, and helped tune the PID controller.

OUTCOME A closed-loop system that holds the pendulum upright and recovers from disturbances.

SolidWorks

Arduino (C++)

PID control

3D printing

SCHOOL MACHINE DESIGN

Two-Stage Gear Reducer

A 9:1 reducer for an industrial conveyor drive, taking 6 hp at 400 rpm down to 45 rpm.

MY ROLE Sized the two-stage gear reduction and ran the bending and moment failure analysis. Designed every part in the assembly except the shaft.

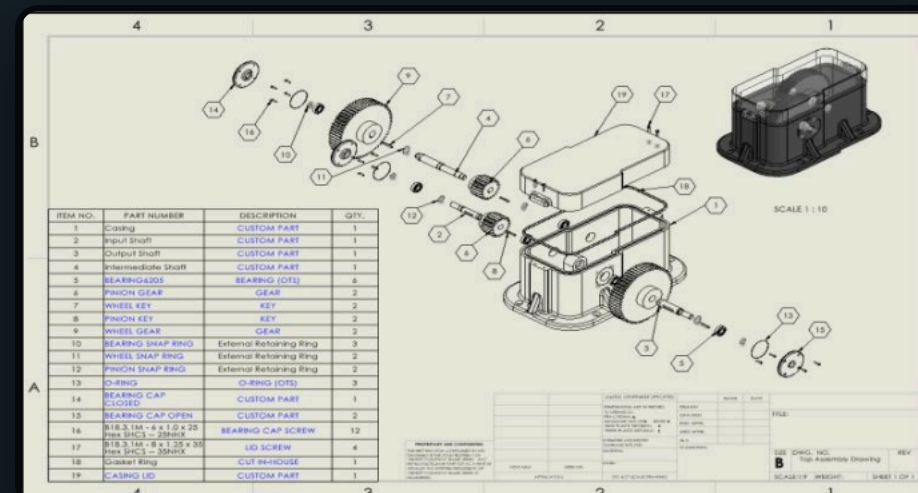
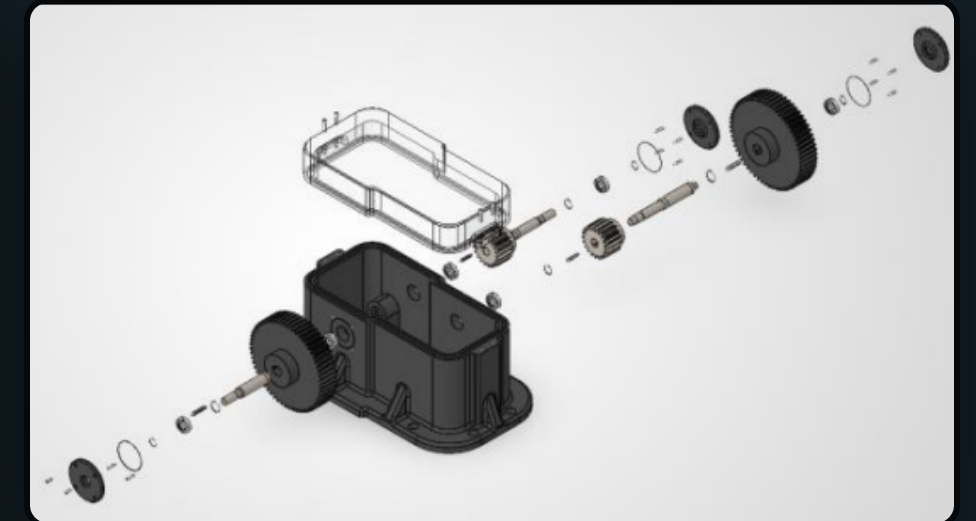
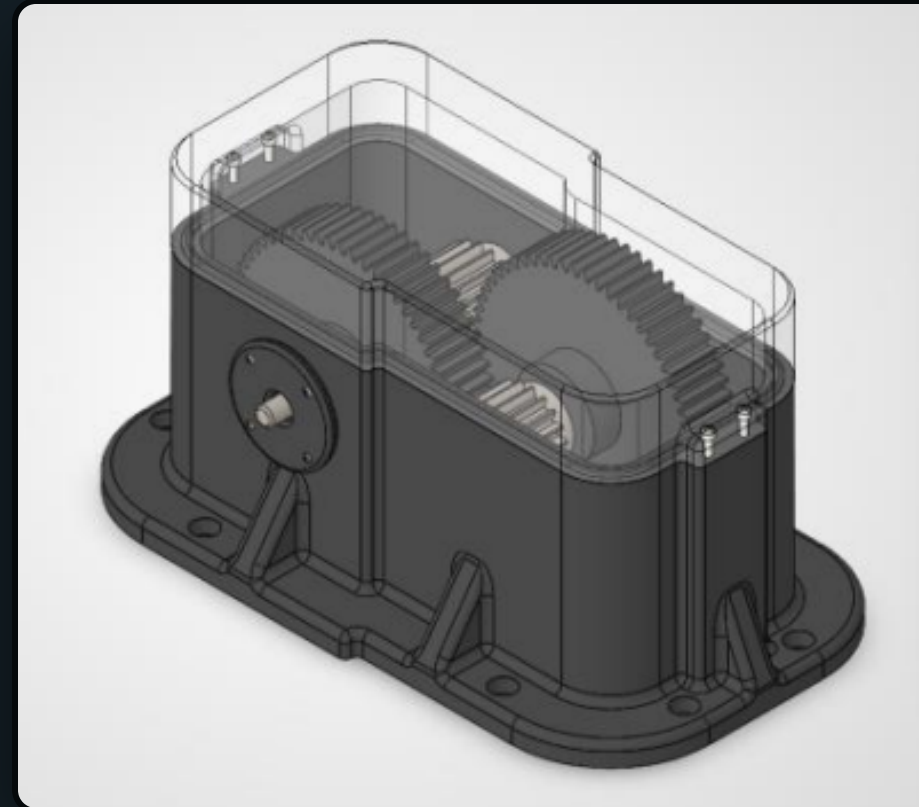
OUTCOME A sealed unit validated for a five-year service life, delivered with a full drawing package.

SolidWorks

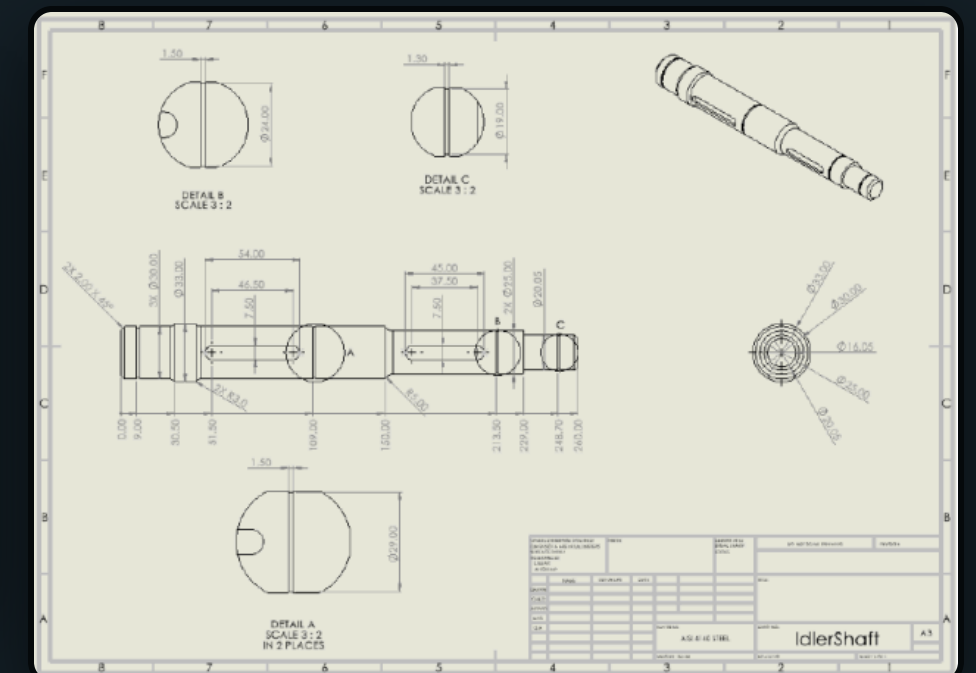
Gear design

Failure analysis

Engineering drawings



ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	Casing	CUSTOM PART	1
2	Input Shaft	CUSTOM PART	1
3	Output Shaft	CUSTOM PART	1
4	Intermediate Shaft	CUSTOM PART	1
5	BEARING (G1)	BEARING (G1)	2
6	BEARING (G2)	BEARING (G2)	2
7	WHEEL KEY	KEY	2
8	WHEEL KEY	KEY	2
9	WHEEL GEAR	GEAR	2
10	BEARING SNAP RING	External Retaining Ring	2
11	BEARING SNAP RING	External Retaining Ring	2
12	BEARING CAP	CUSTOM PART	1
13	BEARING CAP	CUSTOM PART	1
14	BEARING CAP OPEN	CUSTOM PART	2
15	BEARING CAP OPEN	CUSTOM PART	2
16	BEARING CAP OPEN	CUSTOM PART	2
17	BEARING CAP OPEN	CUSTOM PART	2
18	BEARING CAP OPEN	CUSTOM PART	2
19	BEARING CAP OPEN	CUSTOM PART	2



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SCHOOL MCG 2101

Turbine Blade Mold Polisher

A pneumatic gantry that polishes 9-metre wind turbine blade molds automatically, replacing a job that used to be slow and hard on people's arms.

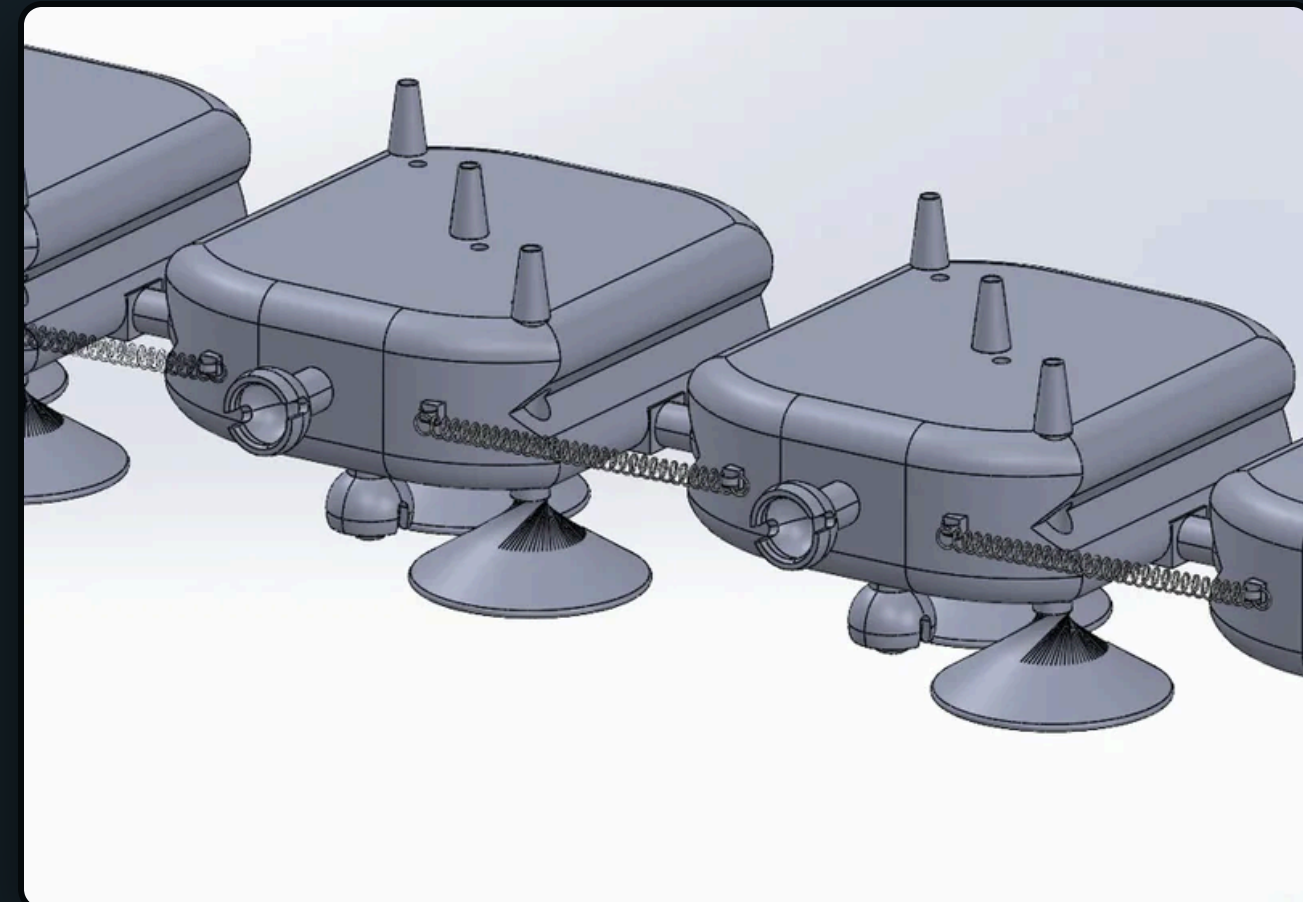
MY ROLE Designed the mechanical layout and motion system: the gantry, the yoke, and the spring-connected plates that follow the curve.

OUTCOME A complete CAD assembly, a full drawing package, and an interactive 3D model that holds constant pressure across the surface.

SolidWorks

Engineering drawings

Pneumatic actuation



THANKS FOR LOOKING

Let's build something

If any of this caught your interest, or you have a project you want a hand with, reach out. I'm always up for collaborating.

chris.alrahi@gmail.com

[LinkedIn](#)

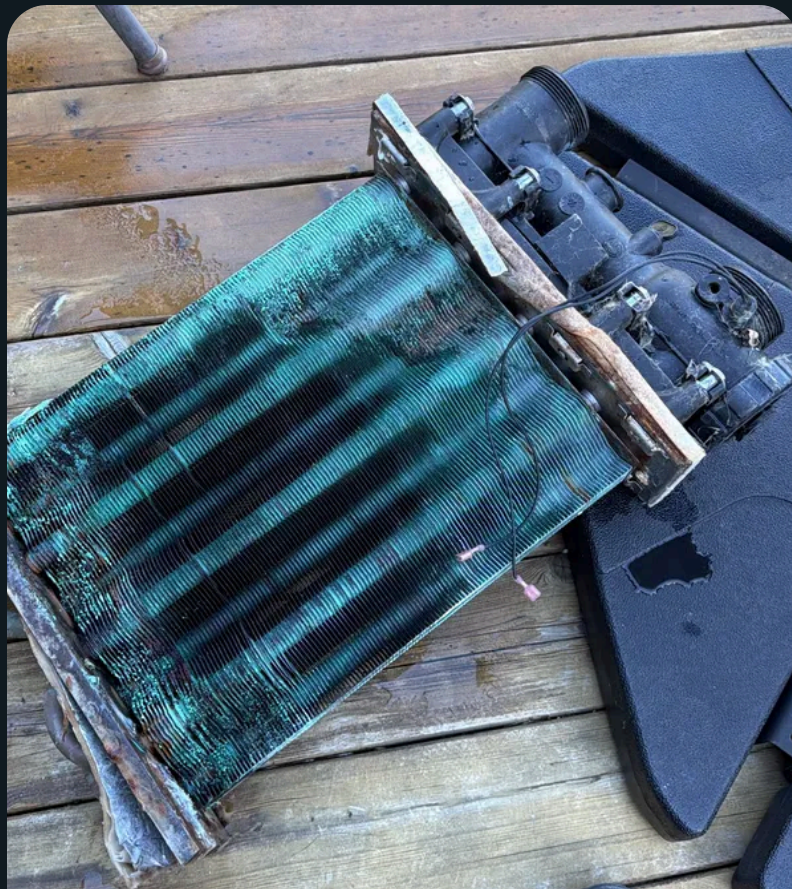
[My Website](#)



HONORABLE MENTIONS

Builds & Repairs

Things I've built or fixed on my own time, mostly because paying someone else to do it felt like a missed opportunity.



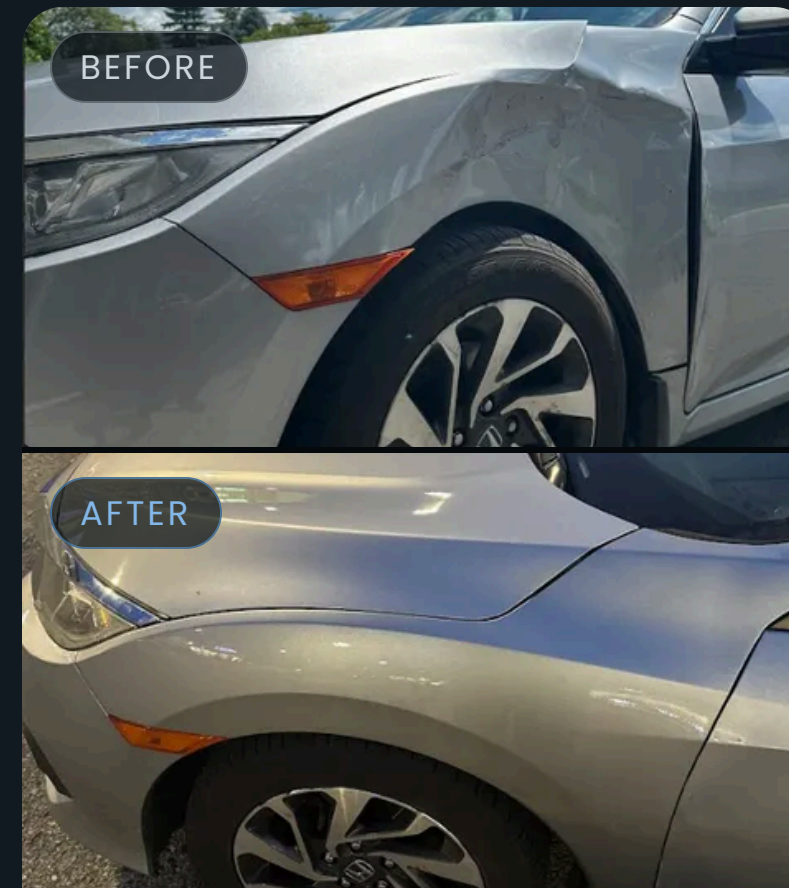
Pool Heater

Found a burst pipe in the heat exchanger, sourced a new one and swapped it in myself.



Hot-Tub Heater

Traced a hot tub error to a burnt heater port, then desoldered and replaced it on the board.



Car Fender Repair

Filled and sanded the cracked panel, matched the paint, refit it myself.



Ping-Pong Top

A fold-away top that turns our pool table into a ping pong table in seconds.



SCHOOL

BEYOND THE PALE BREWING

Beer Goggles

A gravity and temperature monitoring system that reads straight from the fermentation tank, so the brewers don't have to sample by hand.

MY ROLE

Led our team of five, handled mechanical integration, sensor calibration, and the Arduino code tying it together.

OUTCOME

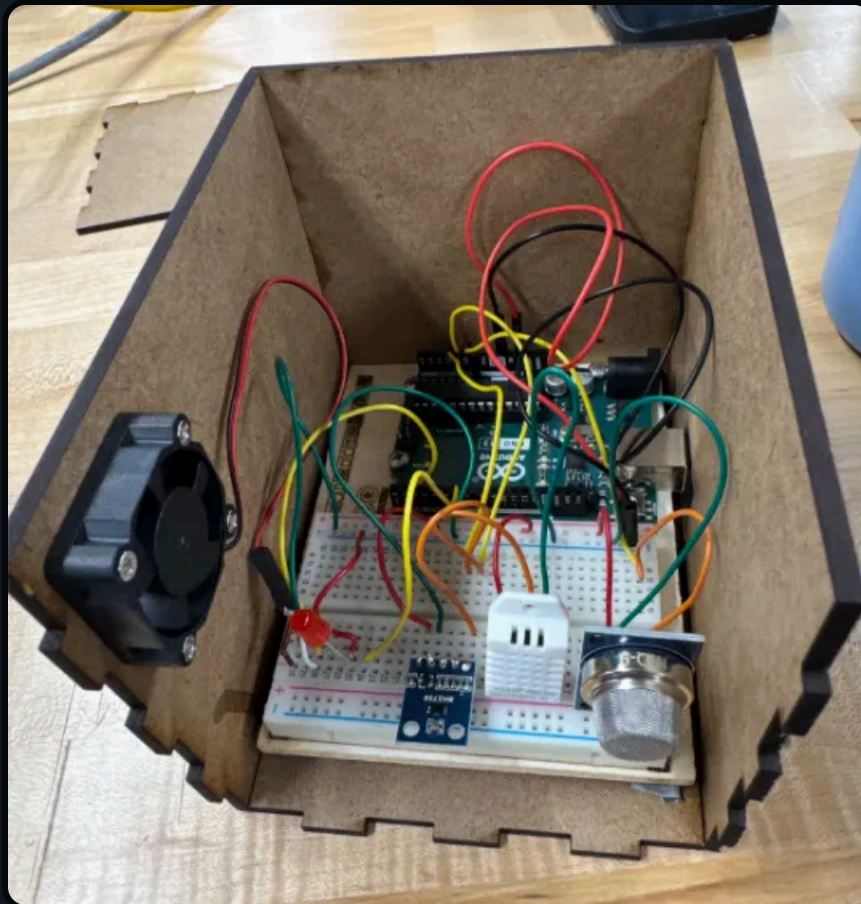
Read specific gravity within ± 0.1 across three rounds of testing, handed off with a full user manual.

Arduino (C++)

Load cell + ultrasonic

SolidWorks

3D printing



SCHOOL EMBEDDED SYSTEMS · ELG 3336

Greenhouse Climate Controller

An automated setup my partner and I built to watch temperature, humidity, light, and air quality, and correct the environment on its own.

MY ROLE We designed it together and I coded the whole thing: reading the sensors, processing the data, switching the fan and lights.

OUTCOME A low-cost, modular controller that held the environment stable with nobody touching it.

Arduino (C++)

DHT22 / BH1750 / MQ-9

Circuit design

